Numerical evaluation of the three-loop $V \to q\bar{q}g$ finite remainders in full colour, in decay and production kinematics

All the full-colour finite remainders are linear combinations, with rational coefficients, of a finite set of pure functions: iterated integrals over a 20-letter alphabet. Evaluating the remainders therefore means evaluating this set of iterated integrals, in the decay region and in the production regions. These notes describe how:

- **Part 1:** the iterated integrals and their numerical evaluation in the decay region;
- **Part 2:** the analytic continuation to the production regions;
- **Part 3:** the assembly of the helicity amplitudes, and the spurious thresholds;
- **Part 4:** the checks, and the precision on the LHC phase space.

The implementation is in `iter int /generators_deq` (evaluator) and `package/` (the shippable package `zgqq_eval`).

Part 1. The iterated integrals and their evaluation

1 Input

- **Variables.** $x = s_{12}/q^2$, $y = s_{13}/q^2$, $z = s_{23}/q^2 = 1 - x - y$. The decay region is the triangle $x, y, z > 0$.
- **Alphabet.** The letters are, with $r = \sqrt{xyz}$:

$$
\begin{array}{llll}
W_1 = x, & W_2 = y, & W_3 = z, & W_4 = 1 - x, \\
W_5 = 1 - y, & W_6 = x + y, & W_7 = \dfrac{xz - ir}{xz + ir}, & W_8 = \dfrac{xy - ir}{xy + ir}, \\
W_9 = 1 - y - xy, & W_{10} = 1 - x - xy, & W_{11} = x - y + y^2, & W_{12} = y - x + x^2, \\
W_{13} = (1 - x)^2 - y, & W_{14} = (x + y)^2 - y, & W_{15} = (x + y)^2 - x, & W_{16} = (1 - y)^2 - x, \\
W_{17} = x + xy + y^2, & W_{18} = x^2 + y + xy, & W_{19} = (1 - y)^2 + xy, & W_{20} = (1 - x)^2 + xy.
\end{array}
$$

  In the decay region the functions involve only W_1–W_8 and W_{11}–W_{16}; the others enter in the production regions (part 2). W_7, W_8 are the odd letters: they contain the square root r.

- **The generators G_k.** The 579 functions G_k of `Zgqq_finite_remainders/Generators.wl`, in terms of which all the remainders are written. Each is a combination with rational coefficients of iterated integrals

$$\Pi[a_1, \ldots, a_n], \qquad d\,\Pi[a_1, \ldots, a_n] = d \log W_{a_n}\, \Pi[a_1, \ldots, a_{n-1}],$$

of weight $n \le 6$. By weight there are $1, 5, 24, 75, 203, 247, 24$ of them.

2 Boundary conditions

The base point is the corner $c_1 = (x, y) = (0, 0)$, with shuffle regularisation: $\mathrm{II}[\,] = 1$ and every non-empty iterated integral vanishes at the base point once its logarithmic divergences are regularised. Near the corner each function is a polynomial in $\log x$, $\log y$ times power series, and the regularised constants are zero. All boundary information is therefore exact. Its values at the other two corners $c_2 = (1, 0)$ and $c_3 = (0, 1)$ follow by exact transport along the edges (GPLs).

3 The differential equation

Word space. A function of weight n is a combination of words with rational coefficients, $f = \sum_{\mathbf{a}} c_{\mathbf{a}} \,\mathrm{II}[\mathbf{a}]$, $\mathbf{a} = (a_1, \ldots, a_n)$. The *word space* of weight n is the vector space with one basis vector for each word $\mathbf{a}$. The function f is identified with its vector of coefficients $(c_{\mathbf{a}})$ there.

- **Why this is allowed.** The $d\log W_i$ are linearly independent, so iterated integrals with different words are linearly independent functions. A linear relation between functions is therefore the same as a linear relation between their coefficient vectors. Questions such as "is this function a combination of those?" become exact linear algebra on rational vectors.
- **Size.** Only the words that actually occur matter: 1872 of the $20^5 = 3.2 \cdot 10^6$ possible words at weight 5, and 9356 of $6.4 \cdot 10^7$ at weight 6.

Example. With $W_1 = x$, $W_2 = y$, $W_3 = z$, the weight-2 function

$$f = \mathrm{II}[1, 2] + 2\,\mathrm{II}[2, 1] - \mathrm{II}[1, 3]$$

is the vector with entries 1, 2, -1 on the words $(1, 2)$, $(2, 1)$, $(1, 3)$, and 0 on all others. Since $d\,\mathrm{II}[a_1, a_2] = d\log W_{a_2}\,\mathrm{II}[a_1]$,

$$df = d\log y\;\mathrm{II}[1] + 2\,d\log x\;\mathrm{II}[2] - d\log z\;\mathrm{II}[1].$$

The coefficient of each $d\log W_i$ is a function of weight 1: here $S_2 f = \mathrm{II}[1]$, $S_1 f = 2\,\mathrm{II}[2]$ and $S_3 f = -\mathrm{II}[1]$. These are the strips defined below.

Differentiation. Each iterated integral is differentiated through its last letter, so collecting the terms of df by the last letter gives

$$df = \sum_{i=1}^{20} d\log W_i \; S_i f, \qquad S_i f = \sum_{\mathbf{a}'} c_{(\mathbf{a}', i)}\,\mathrm{II}[\mathbf{a}'].$$

In words, $S_i f$ keeps the words of f that end in the letter i and deletes that last letter. It is a function of weight $n - 1$, the coefficient of $d\log W_i$ in df; we call it the *strip* of f along i. In word space S_i is a matrix with entries 0 and 1, which sends the word $(\mathbf{a}', i)$ to $\mathbf{a}'$ and every other word to zero.

Closure: the total generators g_k. To integrate the G_k we need every function that appears in their derivatives, and in the derivatives of those, down to weight 0. So we define, from the top down,

$$V_6 = \mathrm{span}\{G_k \text{ of weight } 6\},$$
$$V_n = \mathrm{span}\{G_k \text{ of weight } n\} + \mathrm{span}\{S_i f :\ f \in V_{n+1},\ i = 1, \ldots, 20\} \qquad (n = 5, \ldots, 1),$$

and $V_0 = \{\text{constants}\}$. By construction S_i maps V_n into V_{n-1}, so the spaces are closed under differentiation. The *total generators* g_k, $k = 1, \dots, 503$, are a basis of these spaces: a maximal set of linearly independent functions among the G_k and the strips, with the G_k taken first.

weight n	0	1	2	3	4	5	6	total
generators G_k	1	5	24	75	203	247	24	579
independent G_k	1	5	24	75	113	245	24	487
new functions from the strips						16		16
total generators g_k: $\dim V_n$	1	5	24	75	113	261	24	503

- **Dependent generators.** The G_k are not linearly independent over $\mathbb{Q}$. Only 487 of the 579 are; at weight 4, for instance, the 203 generators span 113 dimensions.
- **The independent G_k alone are not closed.** The strips of the weight-6 generators contain 16 weight-5 functions that are not combinations of any G_k. Without them the derivatives could not be expressed in terms of the unknowns.

So the system to integrate is that of the 503 total generators g_k. Each G_k is a rational combination of the g_k. The choice of basis of the spaces V_n does not change the count, only the numerics (below). All of this is exact linear algebra in word space, with rational coefficients.

The orthonormal system (double precision). For each $n \le 5$ we choose an orthonormal basis of V_n in word space, where two different words are orthonormal: a matrix Q_n whose columns are coefficient vectors, with $Q_n^T Q_n = 1$. The unknowns are the corresponding functions, the *states*

$$\phi_n = Q_n^T \,\mathrm{II}_n \qquad (\mathrm{II}_n : \text{ the vector of the words of weight } n),$$

479 of them for $n \le 5$. They span the same spaces as the g_k of weight ≤ 5. The 24 weight-6 functions never appear in any derivative (there is no weight 7), so they need no equation of their own: a single quadrature along the path gives them (the read-off below). The derivatives of the states follow from the strips. $S_i Q_n$ lies in V_{n-1}, the span of Q_{n-1}, so $S_i Q_n = Q_{n-1}\,(Q_{n-1}^T S_i Q_n)$, and

$$d\phi_n = \sum_i d\log W_i \; A_i^{(n)}\, \phi_{n-1}, \qquad A_i^{(n)} = (Q_{n-1}^T S_i Q_n)^T.$$

Every entry satisfies $|A| \le 1$: S_i only moves words, and Q has orthonormal columns. The generators are read off along the integration from their own strips, $dG_k = \sum_i d\log W_i \left(R_i^{(n)} \phi_{n-1}\right)_k$ for G_k of weight n, with $R_i^{(n)}$ the coefficients of $S_i G_k$ in the basis Q_{n-1}.

Example. Let a weight-2 space be spanned by

$$f_1 = \mathrm{II}[1,2] + \mathrm{II}[2,1], \qquad f_2 = \mathrm{II}[1,2] - \mathrm{II}[2,1] + \mathrm{II}[1,3].$$

- **The orthonormal basis.** On the words $(1,2), (2,1), (1,3)$ the coefficient vectors are $f_1 = (1,1,0)$ and $f_2 = (1,-1,1)$. They happen to be orthogonal already, so normalising them gives $Q_2 = \big(\frac{1}{\sqrt{2}}(1,1,0),\ \frac{1}{\sqrt{3}}(1,-1,1)\big)$. In general Q_n comes from a QR decomposition of the vectors of the g_k.
- **The states and their equations.** The states are $\phi_{2,1} = f_1/\sqrt{2}$ and $\phi_{2,2} = f_2/\sqrt{3}$. At weight 1 the words are already orthonormal, so $\phi_1 = (\mathrm{II}[1], \mathrm{II}[2], \dots)$. The strips give

  $$d\phi_{2,1} = \tfrac{1}{\sqrt{2}}\big(d\log y\ \mathrm{II}[1] + d\log x\ \mathrm{II}[2]\big), \qquad d\phi_{2,2} = \tfrac{1}{\sqrt{3}}\big(d\log y\ \mathrm{II}[1] - d\log x\ \mathrm{II}[2] + d\log z\ \mathrm{II}[1]\big),$$

  i.e. the matrices $A_i^{(2)}$ have entries $\pm 1/\sqrt{2}$, $\pm 1/\sqrt{3}$ and 0, all of size at most 1.

- **A badly scaled basis for comparison.** The pair f_1, $f_2' = f_2 + 1000\, f_1$ spans the same space. But the equation of f_2' has coefficients ~ 1000. Recovering $f_2 = f_2' - 1000\, f_1$ from the unknowns cancels two terms of size ~ 1000, so three digits are lost; the integration errors are amplified the same way. With the g_k of the real system such effects reach many orders of magnitude. The orthonormal basis avoids them by construction.

The integer system (exact, quad precision). For exact work we use a basis over the integers of the same spaces (HNF and LLL), in which every connection matrix M_i is integer:

$$dF = \sum_i d\log W_i \; M_i F.$$

At weights 0–4 and 6 it is a true integral basis with the same counts. At weight 5 a true integral basis needs huge integers, so we keep a redundant spanning set instead: the 247 generators and 97 strips of the weight-6 generators, 344 functions for 261 dimensions. That makes 586 functions in total, and the entries are at most $3.5 \cdot 10^5$. The G_k are integer combinations of the 586 functions F, and the states ϕ_n are exact rational combinations of them, so values carry over exactly between the two systems.

Why the orthonormal system.

- **A basis is needed.** The generators are linearly dependent, so they cannot be integrated directly as a system: the unknowns must be a basis of the spaces V_n, e.g. the total generators g_k.
- **A naive basis is badly conditioned.** The natural choice of unknowns, the g_k themselves, gives connection matrices with rational entries of about 100 digits. The integer basis brings them down to integers, but still up to $3.5 \cdot 10^5$, while its functions reach $4.5 \cdot 10^4$. The derivatives are then sums of large terms that cancel, so every step of the integration loses digits.
- **The orthonormal basis is well conditioned.** With Q_n orthonormal in word space, $A = Q^T S Q$ has all entries $|A| \le 1$, since S only moves words, and the states ϕ_n stay of order one ($|\phi| \le 3$ at P_0). In double precision this gives the best accuracy: at the same points the median error is $2 \cdot 10^{-12}$, against 10^{-11} for the integer system integrated in double. It also uses the minimal number of functions: 479 states plus the 24 weight-6 read-offs.
- **The price.** Q_n comes from a floating-point orthonormalisation, so the matrices A are exact only to double precision. This system cannot go beyond double, and its matrix rounding is the systematic error discussed in section 6. For quad precision and for the exact series we use the integer system.

4 The reference point P_0 of the decay triangle

All evaluations start from one reference point at the centre of the triangle, $P_0 = (0.3, 0.3)$, where the 586 functions are stored once and for all. They are determined to high precision as follows.

1. A double series at the corner. From the exact series of the blow-up charts at c_1, chart 1 ($x = ty^2$) and its mirror ($y = tx^2$), each to order t^{16}, B^{48}, we undo the blow-up ($t = x/y^2$, $\log t = \log x - 2\log y$). The result is a genuine double series,

$$F(x,y) = \sum c\, x^{a/2}\, y^{b/2}\, \log^k x\, \log^l y, \qquad c \text{ exact rational (times 1 or } i).$$

No negative powers appear, and the two charts agree exactly on all $1.3 \cdot 10^6$ terms they share. Several letters vanish at c_1 ($x + y$, $x - y + y^2$, $(x+y)^2 - x$, …), but the functions do not see them there.

The union of the two charts holds every term of total degree $\frac{a+b}{2} \le 32$, so the truncation error is $\approx (|x|+|y|)^{33}$.

2. Two start points, $S_A = (0.05, 0.05)$ and $S_B = (0.04, 0.06)$, evaluated from the series at 40 digits (truncation 10^{-34}).

3. Quad transport over two paths. From each start point we integrate the integer system to P_0 along its own complex-deformed straight path:

$$x(t) = S_x + \big(t + 4i\,\delta_x\, t(1-t)\big)(0.3 - S_x), \qquad y(t) = S_y + \big(t + 4i\,\delta_y\, t(1-t)\big)(0.3 - S_y),$$

with $\delta_A = (-0.15, -0.25)$ (below the real plane) and $\delta_B = (0.1, 0.2)$ (above). The root r is integrated along $(dr = r\, d\log(xyz)/2)$.

- The integrator is Bulirsch–Stoer (modified midpoint, polynomial extrapolation) in double-double arithmetic (≈ 32 digits), with tolerance 10^{-32} relative to the largest function.
- Two points matter. Each half of a double-double number must be read back as a double. And the extrapolation weights $m_l^2/(m_j^2 - m_l^2)$ (m_j the numbers of substeps) must be exact in double-double; in double they cap the accuracy at 10^{-24}.

Result. The two determinations agree to $3 \cdot 10^{-28}$ relative to the largest function ($4.5 \cdot 10^4$): 27–28 digits. Further start points ($(0.06, 0.06)$, $(0.06, 0.04)$, $(0.08, 0.08)$) and a second path from S_A agree within $2 \cdot 10^{-27}$. The values are mapped exactly onto the states ϕ_n of the double system.

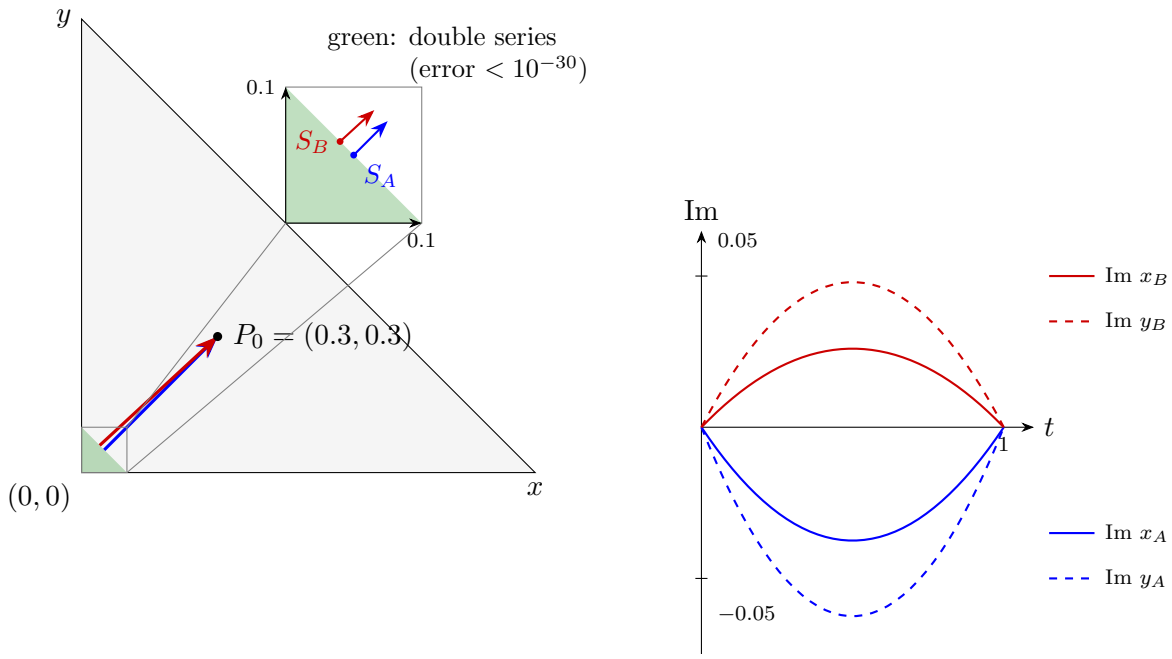


Figure 1. The construction of P_0. Left: the two paths (real part) from the series region to P_0, with a zoom of the corner. Right: the imaginary parts along the paths, $4\delta\, t(1-t)(0.3 - S)$.

5 Transport to any point

A point P of the triangle is reached from P_0 along a complex-deformed straight path, $z(t) = P_0 + \big(t + 4i\delta\, t(1-t)\big)(P - P_0)$ for each coordinate.

- **The deformation.** The letters vanish on curves, and their complex zeros can lie on a straight path. Along the path every letter is a polynomial of degree ≤ 4 in t, so its zeros are found as roots. We use the first deformation from a fixed list whose path keeps a distance $\geq 10^{-3}$ (in t) from every zero.
- **Double.** The orthonormal system is integrated by Bulirsch–Stoer (`cpp/tridents_bs`). This takes about 0.07 s per point.
- **Quad.** The integer system is integrated in double-double (`cpp/iter_quad`), at 5–10 s per point.

The method works anywhere inside the triangle. Close to its edges, where the remainders approach their unresolved (soft, collinear, high-energy) limits, double precision loses digits (section 6). There, quad or the corner series are the right tools.

6 Assessing the precision

Where digits are lost. The remainder is a large sum of rational functions times generators (part 3). Near the edges the rational functions grow (like $1/x^3$, or $(x+y)^{-5}$ at the corners) while the generators grow only logarithmically, so the terms cancel. The sum of |terms| exceeds the value by $\sim 10^4$ in the bulk and up to 10^9 near the edges. Double keeps $16 - \log_{10}(\text{cancellation})$ digits; quad keeps 16 more.

Method 1: double against quad. The same point is evaluated in double and in quad. The difference is the actual error of double, since quad is good to $\sim 10^{-23}$ in the bulk. This is the reference test, used for all the precision statements below.

Method 2: double on two paths (the built-in estimate). Every point is evaluated twice in double, on two different deformed paths, and the estimate is

$$\varepsilon_{\text{est}} = 10\,|R_1 - R_2| + \epsilon \sum |\text{terms}|, \qquad \epsilon = 10^{-13}.$$

The path difference catches the path-dependent integration error, but not what both paths share:

- **The start values.** With the old double-precision P_0, whose errors reached $5 \cdot 10^{-8}$ absolute, the estimate missed values in the subleading colour layers. The quad P_0 removed this.
- **The systematic error of the double system.** Its matrices are floating point. The generator errors of the two paths are 85% correlated, and changing the integration tolerance from 10^{-11} to 10^{-15} does not change them. Amplified by the cancellation, this is covered by the term $\epsilon \sum |\text{terms}|$.

Against method 1 the estimate was an upper bound in every test, but about 1000 times too large. A sharper option is an independent second evaluation with the integer system in double: it bounds the error to ~ 40 times, with 3 misses in 924.

Part 2. Analytic continuation to the production regions

7 The new variables and the new triangle

In a production region one invariant exceeds q^2 and the other two are negative. We take $z > 1$ as the example; it is the channel with partons 2 and 3 incoming. With

$$v = \frac{1}{z}, \qquad u = -\frac{y}{z}, \qquad w = -\frac{x}{z} = 1 - u - v, \qquad u, v, w > 0,$$

the region becomes a triangle again. $v = q^2/\hat{s}$ is the inverse energy: $v \to 1$ is threshold and $v \to 0$ is high energy. In the code the region's triangle is written $(X, Y) = (w, u)$ with $Z = v$, so $(x, y) = (-X/Z, -Y/Z)$. The other regions are analogous:

$$x > 1 : \ (x, y, z) = (1/Z, -X/Z, -Y/Z), \qquad y > 1 : \ (x, y, z) = (-X/Z, 1/Z, -Y/Z).$$

8 The differential equation in the new variables

Substituting the map into the twenty letters, every letter becomes a constant times a power of Z times a letter of the same alphabet in (X, Y): *the alphabet closes.* For $z > 1$:

W_1	W_2	W_3	W_4	W_5	W_6	W_{11}	W_{12}	W_{13}	W_{14}	W_{15}	W_{16}
W_1'/Z	W_2'/Z	$1/Z$	W_5'/Z	W_4'/Z	W_6'/Z	W_{12}'/Z^2	W_{11}'/Z^2	W_9'/Z^2	W_{18}'/Z^2	W_{17}'/Z^2	W_{10}'/Z^2

So $d\log W_i = \sum_j c_{ij}\, d\log W_j'$ with integer c_{ij}, and the region carries the same kind of system:

$$dF = \sum_j d\log W_j'\ M_j' F, \qquad M_j' = \sum_i c_{ij} M_i \quad (\text{and } A_j'^{(n)} = \sum_i c_{ij} A_i^{(n)} \text{ for the orthonormal system}).$$

The same integrators apply unchanged. In the region the letters $W_9, W_{10}, W_{17}, W_{18}$ enter, but their zeros lie outside the region's triangle.

The square root and its Galois symmetry. The odd letters $W_{7,8} = (a - ir)/(a + ir)$ contain $r = \sqrt{xyz}$.

- **The symmetry.** Under the Galois conjugation $r \to -r$, $W_{7,8} \to 1/W_{7,8}$, so their $d\log$ change sign. The functions split into even and odd parts, and the odd ones enter the remainders multiplied by i/r (one of the transcendental multipliers). Each remainder is therefore invariant, provided the same branch of r is used in the letters and in the prefactor.

- **The continued branch.** In the region $xyz > 0$ again, but the continuation fixes the branch. With $\log x \to \log|x| + i\pi$ and $\log y \to \log|y| + i\pi$ (next section), $\sqrt{x}\sqrt{y} \to (i\sqrt{|x|})(i\sqrt{|y|})$, so

$$r \to -\sqrt{|xyz|} = -\frac{\sqrt{XYZ}}{Z^2}.$$

- **In the region's variables.** Expressed through the region's own root $R = \sqrt{XYZ}$, the odd letters map to $d\log W_7 \to d\log W_7' + d\log W_8'$ and $d\log W_8 \to -d\log W_8'$. The rational prefactors are evaluated with $r = -\sqrt{|xyz|}$.

9 The continued boundary values

The corner $X, Y \to 0$ of the region's triangle (threshold) is a decay corner approached from the other side: c_1 for $z > 1$, c_2 for $x > 1$, c_3 for $y > 1$. Its two small variables are $(-X/Z, -Y/Z)$, i.e. (x, y) for $z > 1$. Both cross zero from above (Feynman prescription, every invariant $+i0$), so the corner series is continued by

$$\log s \to \log|s| + i\pi, \qquad \sqrt{s} \to i\sqrt{|s|} \qquad \text{for each small variable } s.$$

In a blow-up chart SMALL $= t\,\mathrm{BIG}^2$ this reads $\log t \to \log|t| - i\pi$, $\log B \to \log|B| + i\pi$.

- $z > 1$ **(corner c_1).** The decay double series, evaluated at $x = -X/Z$, $y = -Y/Z$ with this continuation.
- $x > 1$, $y > 1$ **(corners c_2, c_3).** Here the functions are *not* double series in the two small variables, because the letter $1 - x = y + z$ (resp. $1 - y = x + z$) vanishes at the corner. Undoing the blow-up leaves negative powers already at weight 1, in about 520 functions. We use the exact series of one chart instead (chart $y = tz^2$ at c_2, $x = tz^2$ at c_3), continued as above. It is evaluated at small $|t| \approx 0.02$, i.e. at start points close to the edge $X = 0$.

The series coefficients are exact. In the decay, the chart series at c_2 and c_3 agree with a quad integration from P_0 to 10^{-27}, which checks their boundary constants.

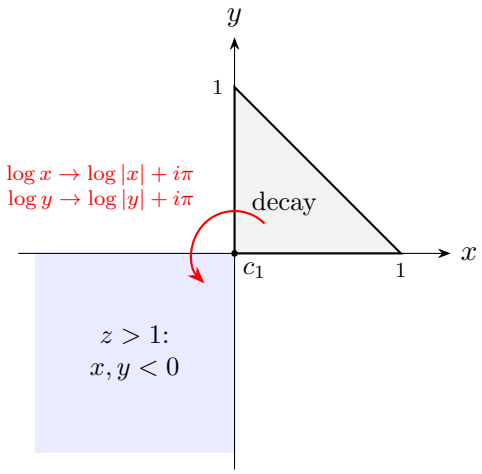


Figure 2. The region $z > 1$ is the quadrant $x, y < 0$ of the plane (x, y); in its own variables it is the triangle $(X, Y) = (w, u)$. It touches the decay triangle at the corner c_1, which is the threshold corner $X, Y \to 0$ of the region. The continuation is the half-turn around c_1, with both small variables crossing zero from above.

10 The reference point of each region

Each region's P_0 is its hub $(X, Y) = (0.3, 0.3)$, determined exactly as in the decay. We take two start points near the region's corner, evaluated from the continued series, and transport each in quad along its own path, with the region's integer system.

region	corner, series	start points (X, Y)	truncation	two paths agree
$z > 1$	c_1, double series	$(0.05, 0.05)$, $(0.04, 0.06)$	10^{-33}	$1.5 \cdot 10^{-27}$
$x > 1$	c_2, chart $y = tz^2$	$(0.001, 0.2)$, $(0.0005, 0.15)$	$2 \cdot 10^{-29}$	$1.9 \cdot 10^{-27}$
$y > 1$	c_3, chart $x = tz^2$	$(0.001, 0.2)$, $(0.0005, 0.15)$	$5 \cdot 10^{-30}$	$2.5 \cdot 10^{-27}$

Inside the region's triangle nothing is crossed, so no $i0$ is needed along these paths.

Cross-check from the decay P_0. The region's P_0 can also be reached by integrating numerically from the decay P_0 to the physical image of the hub. The path is deformed so that the two variables that become negative cross zero with a positive imaginary part, and the root arrives at $-\sqrt{|xyz|}$. This direct transport agrees with the series-based values to $7.8 \cdot 10^{-28}$ (z), $7.9 \cdot 10^{-28}$ (x) and $1.7 \cdot 10^{-27}$ (y). It is used only as a check, not as the definition.

Part 3. The helicity amplitudes

11 Assembly

Each finite remainder (coefficient $\alpha_{1,2,3}$, $\gamma_{1,2,3}$, colour layer $N_c^a N_f^b N_{f,Z}^c$, 84 in total) is

$$\mathcal{R} = \sum_{r,k,m} q_{rkm}\, \mathcal{R}_r(y,z)\, G_k\, c_m,$$

with 129 rational functions $\mathcal{R}_r$, the 579 generators G_k, 15 transcendental multipliers c_m (powers of π, ζ_3, ζ_5, i, and i/r), and rational numbers q (`Zgqq_finite_remainders`).

- **Double.** The rows are merged per (rational function, generator, power of r) and summed as matrix products: 2 ms per point.
- **Quad.** The same sums, with the rational functions and the coefficients in double-double: 21 ms per point. A mixed scheme (double generators, quad sums) gains at most a factor of a few, because the loss is in the generators themselves.
- **Production regions.** The generators come from the region's system. The rational functions are evaluated at the physical point (x, y), the image of (X, Y), with $r = -\sqrt{|xyz|}$.

The result is the decay coefficient analytically continued into the region. The crossing itself (relabelling momenta and helicities, spinor phases) is applied afterwards.

12 Spurious thresholds: the quadratic letters

The quadratic letters $W_{11}, \ldots, W_{16}$ vanish on curves *inside* the decay triangle (figure 3). These are spurious thresholds: the amplitude is smooth across them. Still, the differential equation has poles there, and the rational functions have poles on W_{15}, W_{16}. The singular behaviour cancels in the remainder, but the evaluation has to deal with it.

- **Paths.** The deformation is chosen so that the path keeps away from the complex zeros of all letters (section 5).
- **End points on or near a curve.** If a point lies closer than NEAR$\cdot\rho$ to a curve ($\rho = \min(x, y, z)$), it is replaced by the symmetric average of $f(P \pm l\, h\, \hat{n})$ across the curve, with $\hat{n}$ the unit normal. In double we use three levels,

  $$f(P) = \tfrac{1}{10}(15\bar{f}_1 - 6\bar{f}_2 + \bar{f}_3) + O(h^6), \qquad \bar{f}_l = \tfrac{1}{2}\big(f(P + lh\hat{n}) + f(P - lh\hat{n})\big),$$

  with $h = \min(10^{-3}, \rho/100)$. In quad, five levels give $O(h^{10})$ at $h = \rho/1000$. NEAR is 10^{-8} for the poles of the differential equation alone and 10^{-5} where the rational functions have poles too (quad: 10^{-12}, 10^{-4}).

- **In the production regions** the spurious curves inside the triangle are those of W'_{11}, W'_{12}. In region x they are also poles of the rational functions (the images of W_{15}, W_{16}).

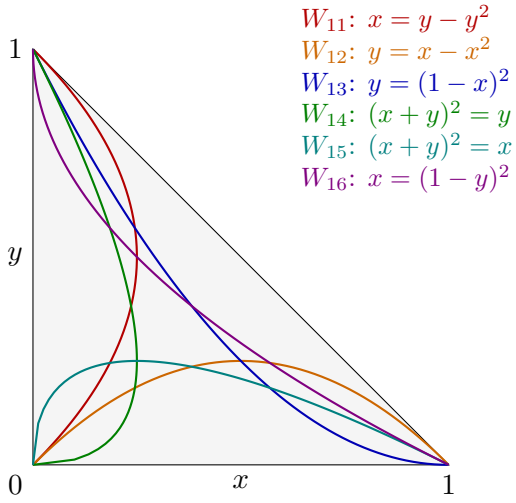


Figure 3. The spurious curves of the quadratic letters in the decay triangle. Each runs into two corners.

Part 4. Checks and precision

13 Leading colour against the exact GPL results

The planar (N_c^3, $N_c^2 N_f$, $N_c N_f^2$) parts of the remainders are known exactly in terms of GPLs (2024 planar ancillary files).

check	points	result
decay, N_c^3 vs `decay.m`, double	406 grid points	median 10^{-13}, worst $7 \cdot 10^{-10}$
decay, N_c^3 vs `decay.m`, quad	3 points	$3 \cdot 10^{-16}$ (reference precision)
region y, planar layers vs `lmL.m`, `lpL.m`	$3 \times 6 \times 3$ values	worst $1.5 \cdot 10^{-11}$

The published values are evaluated with GiNaC at 25 digits. Region y is the channel of the published production results, so this test also checks the analytic continuation and the branch of the root against an independent computation.

Further checks, all colour layers.

- In each region at one point, all 84 values, real and imaginary parts separately. The region evaluator agrees with a direct quad integration from the decay to 10^{-28}–$5 \cdot 10^{-23}$. It agrees with the independent continuation through the corner (double) to 10^{-13}–$2 \cdot 10^{-8}$, the accuracy of that method. Every layer has a sizeable imaginary part.

- Double against quad at random points of the decay triangle, all 84 values: median actual error $2 \cdot 10^{-12}$ in the bulk.

14 Precision on the LHC phase space

The domain. In a production region (e.g. $z > 1$) with jet energy E_j and angle θ in the partonic frame, $u = E_j(1+\cos\theta)/\sqrt{\hat{s}}$ and $w = E_j(1-\cos\theta)/\sqrt{\hat{s}}$, so

$$\frac{p_T^2}{M_Z^2} = \frac{u\,w}{v}, \qquad \frac{u}{w} = e^{2\eta^*}, \qquad \sqrt{\hat{s}} = \frac{M_Z}{\sqrt{v}}.$$

Threshold ($v \to 1$) and the collinear edges ($u, w \to 0$) have $p_T \to 0$. We take $15\,\text{GeV} < p_T < 2\,\text{TeV}$, $\sqrt{\hat{s}} \le 13.6$ TeV (the collider energy) and the jet acceptance $|\eta^*| \le 4.5$. In (u, v), with $\kappa_1 = (15/91.19)^2 = 0.0271$ and $\kappa_2 = (2000/91.19)^2 = 481$, this is

$$\kappa_1 v \le u\,w \le \kappa_2 v, \qquad v \ge 4.5 \cdot 10^{-5}, \qquad e^{-9} \le u/w \le e^{9}.$$

It implies $v \le 0.72$ and $u, w \ge 1.2 \cdot 10^{-4}$.

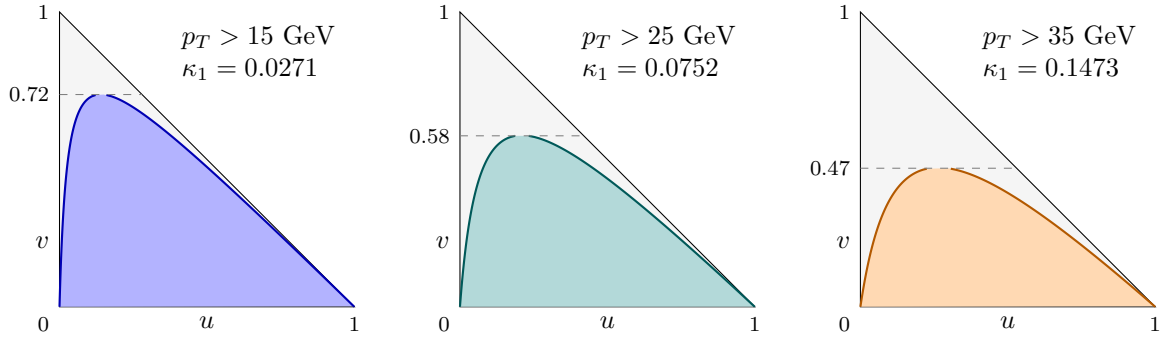


Figure 4. The domain $u\,w \ge \kappa_1 v$ for $p_{T,\min} = 15$, 25 and 35 GeV; the largest v is dashed. The energy and rapidity cuts act at the 10^{-4} scale near the corners, where they cannot be seen on this scale.

The test (`accuracy_study/pheno_domain_check.py`).

- **Points:** about 330 per region, log-spaced in v and dense towards both collinear edges.
- **Double:** every point, with its estimate (method 2).
- **Quad:** the three worst points (by estimate) of each band of $\min(u, w)$ in each region, giving the actual error (method 1).
- **Values:** all 84, i.e. all six helicity coefficients and all colour layers. We quote the colour-summed coefficients ($N_c = 3$, $N_f = 5$, $N_{f,Z} = 1$, the worst of the six) through the bound $\sum_l |c_l|\,|\delta_l| / |\sum_l c_l a_l|$.

Results. Error of the colour-summed coefficients in double, at the checked points of all three regions:

$\min(u, w)$	without the rapidity cut	with $\lvert\eta^*\rvert \le 4.5$
$\ge 10^{-3}$	10^{-8} – $6 \cdot 10^{-6}$	median $1.2 \cdot 10^{-6}$, worst $3.4 \cdot 10^{-6}$
10^{-4} – 10^{-3}	$4 \cdot 10^{-7}$ – $9 \cdot 10^{-6}$	median $1.1 \cdot 10^{-6}$, worst $8.7 \cdot 10^{-6}$
$2 \cdot 10^{-5}$ – 10^{-4}	$3 \cdot 10^{-5}$ – 10^{-4}	outside the domain
$< 2 \cdot 10^{-5}$	10^{-3} – $O(1)$	outside the domain

- **The failing corner.** Without a rapidity cut, the failing points are jets with $\sqrt{\hat{s}} \gtrsim 100\,p_T$ ($\eta^* \gtrsim 4.6$), outside the detector acceptance.

- **Individual layers.** Some single colour layers have large relative errors where they are 10^4–10^6 times smaller than the others. Relative to the largest value at the point the error is at most $4 \cdot 10^{-6}$.

Conclusion. On the LHC domain the double evaluator gives the colour-summed coefficients to better than 10^{-5} in all three regions, typically 10^{-6}. The worst points are at the forward edge of the acceptance and at the highest p_T and energy. A strict 10^{-6} everywhere needs quad at those points. The statement rests on the points checked in quad. Running quad at every point whose estimate exceeds 10^{-6} (1–2 hours) would make it airtight.

Appendix: where things are

`iter int /generators_deq/iisys.py`	closure, orthonormal system
`integral_basis.py`, `out/basis.wl`, `out/deq.wl`	integer system (586 functions)
`quad_data.py`	double series and chart series at the corners; quad runs
`cpp/tridents_bs.cpp`, `cpp/iter_quad.cpp`	integrators: double, double-double
`quad_p0_to_double.py`	exact map from the 586 functions F to the states ϕ_n
`region_p0.py`, `region_quad.py`	region P_0 from the series; direct transport (cross-check)
`gen_eval.py`, `remainder_eval.py`	double evaluation, path choice, spurious averaging, estimate
`quad_eval.py`, `remainder_quad.py`, `ddnp.py`	quad evaluation
`region_lc_check.py`, `region_continuation_check.py`	checks against published results; continuation
`accuracy_study/pheno_domain_check.py`	precision on the LHC domain
`package/`	the shippable package `zgqq_eval`